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Project Title

Smart Moisture Advisor (SMA)

An AI-driven irrigation advisory system for sugarcane

Members

Kartik Verma

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Guide: Dr. Nilesh Yadav

Department of Computer Engineering 2026-27

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Outline of Presentation

- Introduction
- Summary / Finding of Literature Survey
- System Design & Architecture
- Proposed Methodology and Implementation
- Application Outputs
- Results
- Conclusion and Future Scope
- References

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Introduction

- SMA supports sugarcane irrigation decisions for 1,000 plots in Mandya, Karnataka.
- Plot vegetation, soil and climate inputs produce irrigation timing, water volume and pump duration.
- The interface links rain-aware advice with fertigation, risk estimates, a map and pump scheduling.
- Seven predictive models learn agronomy-generated labels. Field outcomes remain unverified.
- Advisories support English, Kannada, Hindi and Marathi, with accept or override feedback.

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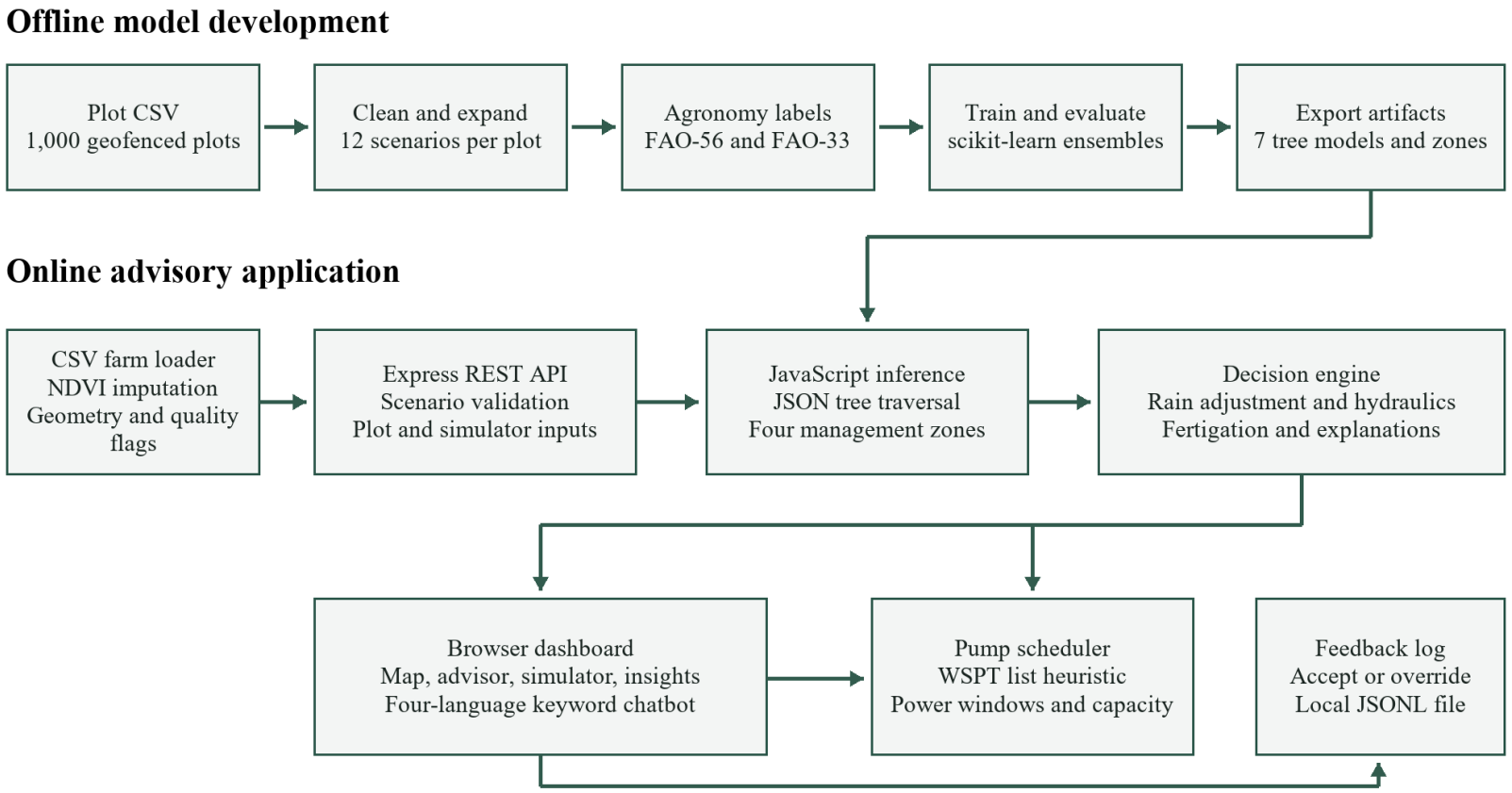
Summary / Finding of Literature Survey

Source	Approach or finding	Gap relative to SMA scope
Allen et al. [1]	FAO crop water balance and ET	Requires suitable local inputs and calibration
Doorenbos and Kassam [2]	Yield response to water deficit	Does not provide an integrated plot interface
Kamienski et al. [3]	SWAMP IoT platform and pilots	Deployment and scaling require infrastructure
Breiman [4] and Friedman [5]	Random Forest and Gradient Boosting	Require representative targets and evaluation
Current project [10]	Integrates advice, maps and scheduling	Independent field validation is still needed

The comparison identifies project requirements, without claiming superiority over the cited systems.

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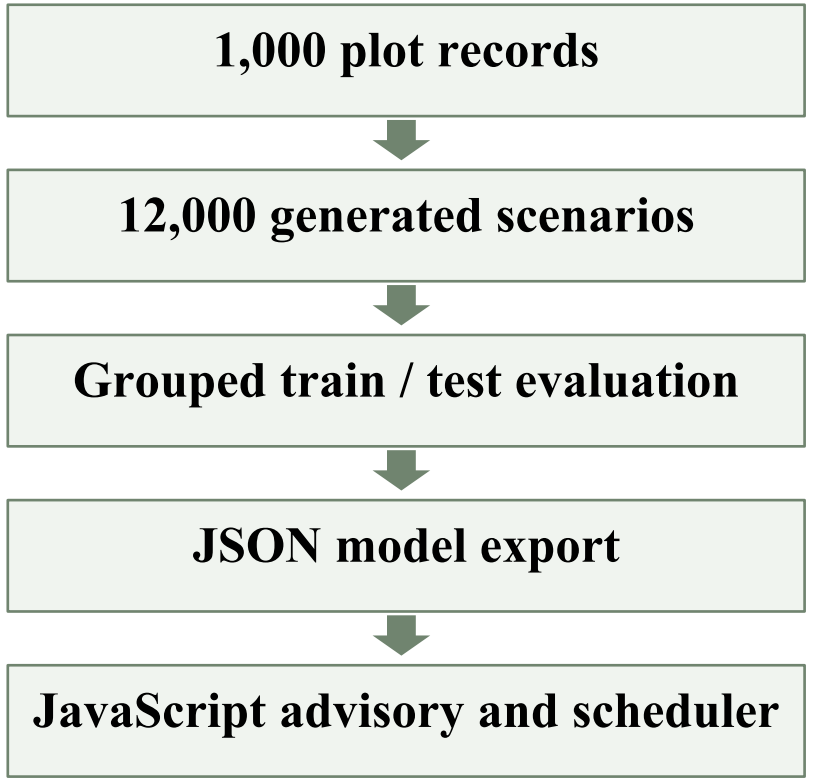
System Design & Architecture



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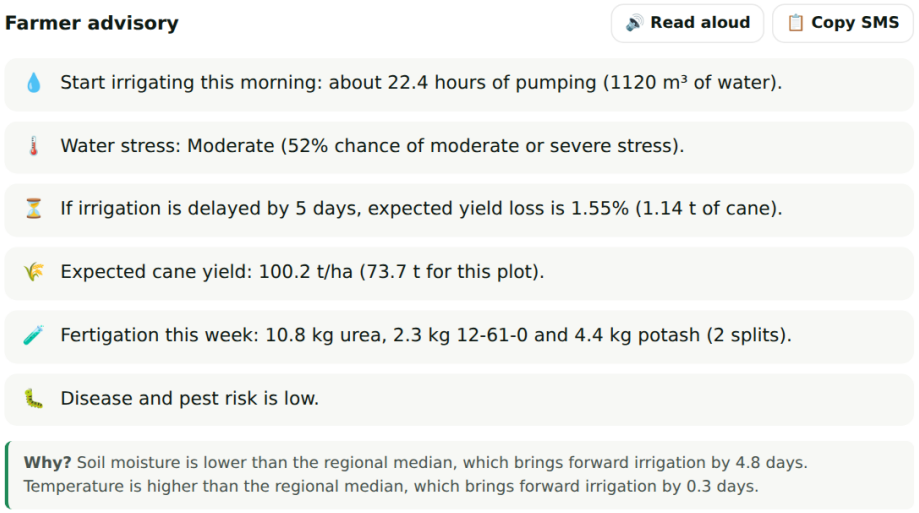
Proposed Methodology and Implementation

- Clean plot data and create twelve scenarios per plot.
- Generate noisy FAO-based labels for irrigation, crop demand and risk.
- Split by plot: 800 training plots and 200 held-out plots.
- Train ensembles, compare linear or logistic baselines, then export JSON.
- Serve predictions through Express and apply hydraulics, fertigation and scheduling rules.

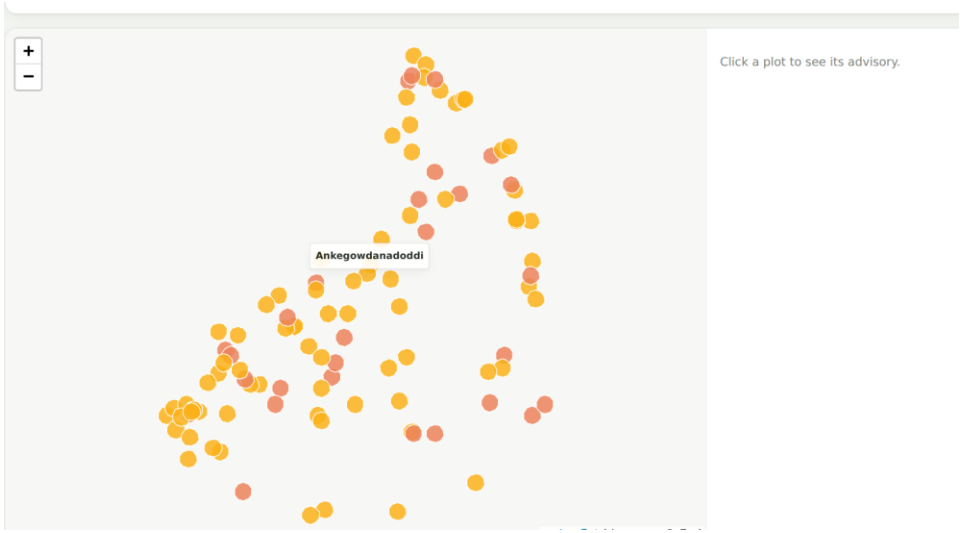


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Application Outputs



Plot advice, yield, fertigation and risk



Geofenced plots coloured by stress

- What-if Simulator recomputes outputs when field conditions change.
- Krishi Mitra uses keyword intent detection in four supported languages.
- Farmers and supervisors can accept or override the recommendation.

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Results

Target	Ensemble score	Baseline score
Irrigation timing	0.9646	0.7444
Crop water requirement	0.9768	0.8843
Irrigation depth	0.9745	0.6206
Cane yield	0.8918	0.5709
Delay loss	0.9755	0.4990
Water stress	87.83%	83.46%
Disease risk	87.29%	81.96%



Hosur schedule
8 plots completed vs 5
5.37 t less estimated loss

R-squared for regression, accuracy for classifiers. Timing MAE: 0.6942 days.
Evaluation uses generated labels on held-out plots. Savings and cane loss are scenario estimates.

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Conclusion and Future Scope

- SMA integrates plot-level prediction, irrigation advice and power-window scheduling in a working web prototype.
- JSON model export supports local JavaScript inference. All 23 existing software tests pass.
- The saved metrics show agreement with generated agronomic labels. Independent field validation is the main next step.
- Future work: measured irrigation and yield records, verified weather feeds, actual planting dates and calibrated uncertainty.
- Operational development: agronomist review, field pilots, role-based access and a reviewed retraining workflow.

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References

[1] Allen et al., Crop Evapotranspiration, FAO Paper 56, 1998. fao.org/4/x0490e/x0490e00.htm

[2] Doorenbos and Kassam, Yield Response to Water, FAO Paper 33, 1979.

[3] Kamienski et al., Smart Water Management Platform, Sensors 19(2), 276, 2019. doi:10.3390/s19020276

[4] Breiman, Random forests, Machine Learning 45, 5-32, 2001. doi:10.1023/A:1010933404324

[5] Friedman, A gradient boosting machine, Annals of Statistics 29(5), 1189-1232, 2001. doi:10.1214/aos/1013203451

[6] Pedregosa et al., Scikit-learn, JMLR 12, 2825-2830, 2011. jmlr.org/papers/v12/pedregosa11a.html

[10] Verma and Soni, Smart Moisture Advisor source and saved outputs. github.com/Interior-Gardener/SMA

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Thank You

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